a:t

Radial Basis Functions (RBF) Kernal Method

	x_1	x_2	y
Train	1	1	2
	3	1	4
Test	2	1	?

test x: $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$$h_\theta(x) = \theta^T \phi(x)$$

$$\theta \in \mathbb{R}^p$$
$$\phi(x) \in \mathbb{R}^p$$

$$h_\theta(x) = \sum_{i=1}^{m} \beta_i \, k(x^{(i)}, x)$$

Let us assume the following

$\beta_1 = 1$
$\beta_2 = 1$ } obtained by Gradient Descent on higher Dimension but (1 and 1) is arbitrary/fake

Kernal = Radial Basis Function (RBF)

$$k(x, z) = \exp(-\|x - z\|^2)$$

where

$$\|x - z\|^2 = (x_1 - z_1)^2 + (x_2 - z_2)^2$$

$$h_\theta(x) = \sum_{i=1}^{n} \beta_i \, k(x^{(i)}, x) = \theta^T \phi(x) = \sum_{k=1}^{n} \beta_i k(x^{(i)}, x)$$

$$= \beta_1 k(x^{(1)}, x) + \beta_2 k(x^{(2)}, x)$$

$$= \beta_1 \exp(-\|x^{(1)} - x\|^2) + \beta_2 \exp(-\|x^{(2)} - x\|^2)$$

since $\|x - y\|^2 = (x_1 - y_1)^2 + (x_2 - y_2)^2$

$$= \beta_1 \exp(-((x_1^{(1)} - x_1)^2 + (x_2^{(1)} - x_2)^2)) + \beta_2 \exp(-((x_1^{(2)} - x_1)^2 + (x_2^{(2)} - x_2)^2))$$

$$= \beta_1 \exp(-((1-2)^2 + (1-1)^2)) + \beta_2 \exp(-((3-2)^2 + (1-1)^2))$$

$$= 1 \exp(-1) + 1 \exp(-1)$$

$$= 0.367 + 0.367 = 0.734$$

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$$h_\theta(x) = 0.734$$